

OBJECT ORIENTED PROGRAMMING With JAVA

Course Code	BCS306A	Course Category:	PLC/IPCC
Teaching Hours/Week (L:T:P)	2:0:2	Total Hours/Semester	32(L) +20(P)
CIE + SEE Marks	50 +50	Semester	III
SEE Duration	3 Hours	Credits	03
Teaching Department: Computer Science & Engineering			
Course Objectives			
1	Introduce the fundamental concepts of Java programming, including data types, variables, arrays, operators, and control statements.		
2	Explain the principles of object-oriented programming by detailing classes, objects, and methods, and demonstrate how to develop basic Java applications using these concepts.		
3	Describe advanced object-oriented features such as inheritance and interfaces, and guide students in developing programs utilizing these concepts.		
4	Illustrate the use of Java packages and exception handling mechanisms, enabling students to construct robust programs that effectively manage errors.		
5	Explore advanced topics like multithreading, enumerations, type wrappers, and autoboxing, providing practical experience through experimentation and application.		
Prerequisites for the Course:			
Basic Programming Knowledge: Understanding fundamental programming concepts such as variables, data types, basic programming constructs.			
Understanding of Programming Languages: Familiarity with preferably procedural languages like C or C++			
Basic Knowledge of Data Structures: Concepts like arrays, lists, stacks and queues			
Mathematical Foundations: Basic mathematical skills, especially logic and reasoning			
Development Environment Setup: Eclipse, IntelliJ IDEA, or NetBeans.			
Problem-Solving Skills: Ability to analyze problems and develop algorithms.			
MODULE-I			6 Hours
An Overview of Java: The Java Buzzwords, How java differs from C and C++ ? Implementing a java program, Java Virtual Machine, Object-Oriented Programming (Two Paradigms, Abstraction, The Three OOP Principles), Using Blocks of Code, Lexical Issues (Whitespace, Identifiers, Literals, Comments, Separators, The Java Keywords).			
Data Types, Variables, and Arrays: Type Conversion and Casting, Automatic Type Promotion in Expressions, Arrays, Introducing Type Inference with Local Variables.			
Operators: Arithmetic Operators, Relational Operators, Boolean Logical Operators, The Assignment Operator, The ? Operator, Operator Precedence, Using Parentheses.			
Control Statements: Java’s Selection Statements (if, The Traditional switch), Iteration Statements (while, do-while, for, The For-Each Version of the for Loop, Local Variable Type Inference in a for Loop, Nested Loops), Jump Statements (Using break, Using continue, return).			
Programming Exercises : Laboratory Experiment list : Q1 and Q2			
Self-Study:. Data Types The Primitive Types (Integers, Floating-Point Types, Characters, Booleans), Variables			
Chapter 1, 2, 3, 4, 5 from text book 1 and Chapter 2 from Ref book1			
MODULE-II			6 Hours
Introducing Classes: Class Fundamentals, Declaring Objects, Assigning Object Reference Variables, Introducing Methods, Constructors, The this Keyword, Garbage Collection.			
Methods and Classes: Overloading Methods, Objects as Parameters, Argument Passing, Returning Objects, Access Control, Understanding static, Introducing final			
Programming Exercises :Laboratory Experiment list : Q3, Q4 and Q5			
Self-Study: Recursion and Introducing Nested and Inner Classes.			
Chapter 6 and 7 from text book 1			
MODULE-III			6 Hours
Inheritance: Inheritance Basics, Using super, Creating a Multilevel Hierarchy, Method Overriding, Dynamic Method			

Dispatch, Using Abstract Classes, Using final with Inheritance, Local Variable, Type Inference and Inheritance, The Object Class. Interfaces: Interfaces, Default Interface Methods, Use static Methods in an Interface, Private Interface Methods. Programming Exercises : Laboratory Experiment list : Q6, Q7, Q8 and Q9 Self-Study: When Constructors Are Executed, Method Overriding.							
Chapters 8, 9 from text book 1							
MODULE-IV							
7 Hours							
Packages: Packages, Packages and Member Access, Importing Packages. Exceptions: Exception-Handling Fundamentals, Exception Types, Uncaught Exceptions, Using try and catch, throw, throws, finally, Java's Built-in Exceptions, Creating Your Own Exception Subclasses, Chained Exceptions. Programming Exercises : Laboratory Experiment list :Q10 and Q11 Self-Study: Multiple catch Clauses and Nested try Statements,							
Chapters 9, 10 from text book 1							
MODULE-V							
7 Hours							
Multithreaded Programming: The Java Thread Model, The Main Thread, Creating a Thread, Creating Multiple Threads, Using isAlive() and join(), Thread Priorities, Synchronization, Serialization and De serialization ,Suspending, Resuming, and Stopping Threads, Obtaining a Thread's State. Enumerations, Type Wrappers and Autoboxing: Enumerations (Enumeration Fundamentals, The values() and valueOf() Methods), Type Wrappers (Character, Boolean, The Numeric Type Wrappers), Autoboxing (Autoboxing and Methods, Autoboxing/Unboxing Occurs in Expressions, Autoboxing/Unboxing Boolean and Character Values). Programming Exercises : Laboratory Experiment list : Q12 and Q13 Self-Study: Interthread Communication							
Chapters 11, 12 from text book 1							
Laboratory Experiment List							
1.	Develop a JAVA program to add TWO matrices of suitable order N (The value of N should be read from command line arguments) using function AddMatrix() .						
2.	Develop a program for multiplying and printing i=3, j=4, k=5 using nested for loops using multidimensional array concept.						
3.	A class called Employee, which models an employee with an ID, name and salary, is designed as shown in the following class diagram. The method raiseSalary(percent) increases the salary by the given percentage and methods setdata() and getdata() to set the data and return the data respectively. Develop the Employee class and suitable main method for printing ID,name and salary by using toString() method. Class diagram of Employee Class <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td style="padding: 5px;">Class name →</td> <td style="padding: 5px; text-align: center;">Employee</td> </tr> <tr> <td style="padding: 5px;">Attribute names →</td> <td style="padding: 5px;"> - Empno : int -Empname : String -salary : int </td> </tr> <tr> <td style="padding: 5px;">Method names →</td> <td style="padding: 5px;"> +setdata(Empno :int,Empname:String, salary:int) : void +getdata() : void + raiseSalary(percent:double) : void + toString() : String </td> </tr> </table> + means public - means private	Class name →	Employee	Attribute names →	- Empno : int -Empname : String -salary : int	Method names →	+setdata(Empno :int,Empname:String, salary:int) : void +getdata() : void + raiseSalary(percent:double) : void + toString() : String
Class name →	Employee						
Attribute names →	- Empno : int -Empname : String -salary : int						
Method names →	+setdata(Empno :int,Empname:String, salary:int) : void +getdata() : void + raiseSalary(percent:double) : void + toString() : String						
4.	Build a Java programming for finding areas of circle, rectangle,triangle(three sides given) using constructor overloading and method overloading techniques . The parameters have to be passed from main to class and hide the data members in the class Develop three objects using parameterized constructor overloading and print the results using toString() .						

5.	Add two objects of feet, inches by passing objects as parameters into a class called Measure with data members feet, inches and parameterized and non parameterized constructors to initialize the data members feet and inches . Use the method Calculate() to add the two objects and print the result object.
6.	Develop a class Employee for implementing multilevel inheritance concept including data fields namely name, salary and a method called setdata() to pass the parameters name and salary from main to class and print name and salary using the method dispdata() and include a non-parameterized constructor to initialize the name to “ ” and salary to zero. Similarly inherit the class SkilledAssist from Employee with an extra attribute allowance and inherit the class Manager from SkilledAsst with an extra attribute bonus . Develop a main Class EmpTest to create objects for the classes Employee, SkilledAsst and Manager to pass the data fields and print them on the console.
7.	Build a Java program for finding interest for the investment from different banks which offer different rate interests such as SBI (8%), ICICI (9%) and AXIS (10%) using the concept of method overriding . Develop a parent class BANK with data members Investment_amount and rate_of_interest initialized to zero. Print results of interests from different banks using different subclasses namely SBI, ICICI and AZIX . The data member Investment_amount has to be passed from main to class and hide the data member in the class. Print the results by invoking the method getInterest() with respective objects in Main class named Override
8.	Develop a JAVA program to create an abstract class Shape with abstract methods calculateArea() and calculatePerimeter() . Create subclasses Circle and Triangle that extend the Shape class and implement the respective methods to calculate the area and perimeter of each shape.
9.	Build a Java program for finding the areas of different shapes like square, rectangle, triangle(3 sides given) with respective parameters. Implement a common interface called Figure which implements three classes namely Sqr, Rect and Tria with parameterized constructors of respective dimensions. Create a reference to the interface and store the objects of the classes into the reference pointer of the interface in the main class.
10.	Create a custom exception called InsufficientBalanceException to represent a situation where a user tries to withdraw more money than is available in their bank account
11.	Create three packages namely Arith, Great and MainPack for storing the modules ArithmeticOperations, GreatestFinder and MainClass respectively. Create different objects for the classes ArithmeticOperations (Add, Sub, Mul, Div) and GreatestFinder (Great2, Great3, Great4) to access the methods which are defined in these classes.
12.	Write a Java program to illustrate five states of the life cycle of the thread (New, Runnable, Running, Non-Runnable (Blocked/Waiting), Terminated) with an example program
13.	Illustrate Real time Application of Data Conversion Using valueOf() in Java with an example program developed for banking account creation application with the class BankAccount and the fields accountHolderName(String), accountNumber(int), balance (double), gender(boolean – true for male and false for female) . Get the inputs through keyboard and print all the bank Information on the console
14.	Write programs to implement the concept of synchronization for printing numbers from 1 to 2000 with and without using synchronization
15.	Write a Java program to serialize and deserialize the Employee object given Q2
Content Beyond the Syllabus	
	Diamond Problem existing in C++ and How it can be solved through Interface in Java with an application program.
Course Outcomes(COs)	
COs	At the end of the course student will be able to
1	Explain the Overview of Java, Data Types, Variables & Arrays and Apply concepts of Operators & Control Statements in Java
2	Infer the concepts of Classes and Methods and Build the object-oriented programs using them.
3	Interpret the concepts of Inheritance and Interfaces and Develop programs using them.
4	Demonstrate the concepts of Packages and Exceptions and Construct the programs for

	analysing them.														
5	Illustrate the concepts of Multithreaded Programming, Enumerations, Type Wrappers & Autoboxing and Experiment with them														
Course Outcomes Mapping with POs & PSOs															
↓ CO / POs→	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	2	2	2	1	3				2	2	1	2	2	2	2
2	2	2	2	2	3				1	2	1	2	2	2	2
3	2	2	2	1	3				1	2	1	3	3	3	3
4	2	2	2	1	3				2	2	1	2	2	2	2
5	2	2	2	1	3				2	2	1	2	2	2	2
CO-Course Outcomes, PO-Program Outcomes, PSO –Program Specific Outcomes 1: Low 2: Medium 3: High															
Course Evaluation System															
Course Type	CIE Marks	Distribution				Description									
IPCC	50	Theory: 25	CIE Test: 20				Average of two tests of 20 marks each								
			Assignment-05				Presentation cum Viva Session (Assignment) : 05								
		Lab: 25	CIE: 15				Execution of all exercises with observation writing on weekly basis: 05 Marks Weekly submission of handwritten Laboratory Record with input and output along with upload of all the experiments on Git repository/e-Box Platform with I/P and O/P: 10 marks								
			IA Test: 10				Minimum One IA test will be conducted after the completion of all experiments.								
Semester End Examinations (SEE) Paper Pattern															
The SEE will be conducted as per Scheduled Timetable announced by Controller of Examinations(CoE)															
1. The SEE question paper contains Ten questions and 2 questions from each Module.															
2. Student shall answer 5 full questions, selecting one full question from each module.															
3. Each question in a module shall contains minimum 2 or maximum 3 sub divisions.															
Question also shall be asked from Laboratory experiments for IPCC Course.															
Text Books:															
1	Java: The Complete Reference, Twelfth Edition, by Herbert Schildt, November 2021, McGraw-Hill, ISBN: 9781260463422														
Reference Books:															
1	Programming with Java, 6th Edition, by E Balagurusamy, Mar-2019, McGraw Hill Education, ISBN: 9789353162337.														
2	Thinking in Java, Fourth Edition, by Bruce Eckel, Prentice Hall, 2006 (https://sd.blackball.lv/library/thinking_in_java_4th_edition.pdf)														
e-Resources															
1	Java Tutorial: https://www.geeksforgeeks.org/java/														
2	Introduction To Programming In Java (by Evan Jones, Adam Marcus and Eugene Wu): https://ocw.mit.edu/courses/6-092-introduction-to-programming-in-java-january-iap-2010/														
3	Java Tutorial: https://www.w3schools.com/java/														